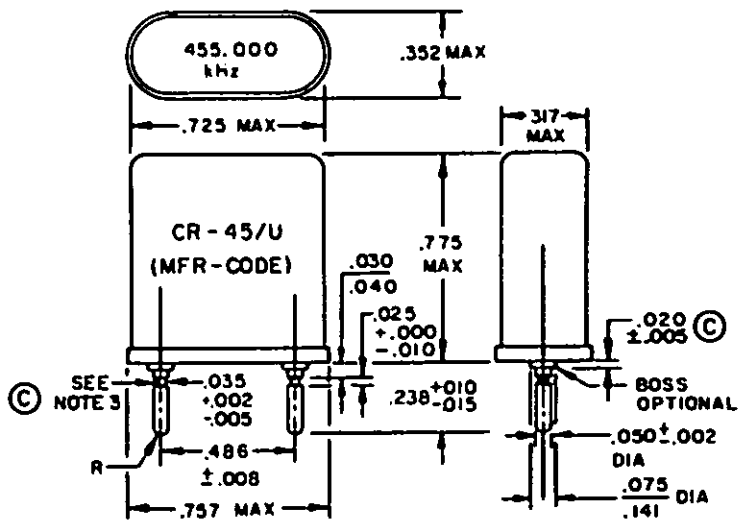
# MILITARY SPECIFICATION SHEET

## CRYSTAL UNIT, QUARTZ, CR-45/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Specification MIL-C-3098.

Pertinent characteristics: 455 kHz; fundamental; noncontrolled; series.




| INCHES | MM | INCHES | MM | INCHES | MM | INCHES | MM |
|---|---|---|---|---|---|---|---|
| .002 | .05 | .020 | .51 | .050 | 1.27 | .352 | 8.94 |
| .005 | .13 | .025 | .64 | .075 | 1.91 | .486 | 12.34 |
| .008 | .20 | .030 | .76 | .141 | 3.58 | .725 | 18.42 |
| .010 | .25 | .035 | .89 | .238 | 6.05 | .757 | 19.23 |
| .015 | .38 | .040 | 1.02 | .317 | 8.05 | .775 | 19.69 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
© 3. The pin undercut may be omitted.

© denotes change

FSC 5955



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TABLE I. Requirements.

| Requirement | Specified value or condition |
|---|---|
| Holder | HC-6/U (MIL-H-10056/2) |
| Dimensions, marking, and configuration | See figure |
| Frequency range | 455 kHz |
| Capacitance, shunt | 5.0 ±2.5 pF |
| Mode of oscillation | Fundamental |
| Government reference standard test set | TS-710/TSM |
| Rated drive level | 2.0 ±0.4 |
| Calibration values: | |
| Equivalent resistance | 2,300 ohms |
| Resistor voltage drop | 2.14 volts |
| Operating temperature range: | -40° to +70° C, inclusive |
| Frequency tolerance 1/ | ±200 PPM |
| Equivalent resistance | 3,300 ohms, maximum |
| Shock (specified pulse): | |
| Frequency change permitted | ±30 PPM |
| Equivalent resistance change permitted | ±15 percent |
| Vibration: 2/ | |
| Frequency change permitted | ±30 PPM |
| Equivalent resistance change permitted | ±15 percent |
| Temperature run: | |
| Frequency change permitted | ±10 PPM |
| Equivalent resistance change permitted | ±15 percent |
| Bond strength | 500 grams, minimum |

1/ There shall be a difference in frequency of 80 ±12 Hertz between operation at series resonance and operation at antiresonance into load capacitance of 32.0 ±0.5 pf. Both measurements for this purpose shall be made at room temperature.

2/ Method 201 of MIL-STD-202, amplitude 0.015 inch (0.030 inch total excursion).

NOTE:

© The socket designed to mate with crystal holder HC-6/U is Part No. M12883/04-01 (See MIL-S-12883/4).
For Navy use, other sockets listed in MIL-STD-242 may be used.

Custodians:
Army - EL
Navy - EC
Air Force - 84

Review activities:
Army - EL, MI, MU, SG, SM
Navy - EC
Air Force - 11, 85
DSA - ES

User activities:
Army -
Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5955-0408-13)



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